

A hand-drawn diagram illustrating the relationship between different programming languages. It consists of three concentric circles. The innermost circle is labeled 'C' and is circled in blue. The middle circle is labeled 'Assembly language'. The outermost circle is labeled 'python' and is circled in blue. An arrow points from the 'python' circle to the top right. Checkmarks are next to 'C++' and 'java' which are also in the outermost circle.

Example 1 : basic CGPA

23 May 2026 11:58

12:24

[illegible]

**Int,
Float,
String,
Char,
Byte,
Unsigned int,
Long int,**

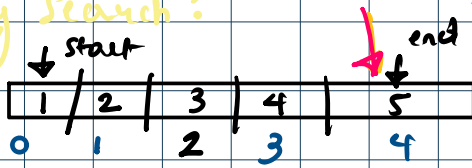
If,
If/else,
Else if,
Switch,

Ternary operator

if / else

ternary operator.

Binary Search:



key = 5

$$\rightarrow \text{mid} = \frac{\text{start} + \text{end}}{2} = 2$$

$(\text{arr}[\text{mid}] > \text{key}) ; (\text{arr}[\text{mid}] < \text{key})$
 $\text{arr}[2] > 5$
 $\text{start} = \text{mid} + 1$

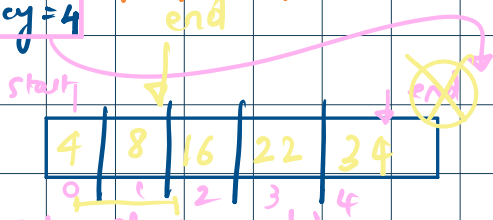
linear search
 $O(n)$

Binary Search
 $O(\log n)$

- English.
- ppt-slides
- Questions
- ↓
- program

4 | 8 | 16 | 22 | 34

key = 4



$(\text{mid} = \frac{\text{start} + \text{end}}{2})$
 $\text{if} (\text{arr}[\text{mid}] == \text{key})$
 return mid;

else if $(\text{arr}[\text{mid}] > \text{key})$
 $\text{arr}[2] > \text{key}$
 $(16 > 4)$
 $\text{end} = \text{mid} - 1$

2
1
0
1

$$\text{mid} = \frac{\text{start} + \text{end}}{2} = \frac{0 + 1}{2} = 0$$

$\text{arr}[\text{mid}] > \text{key}$

0 | 1 | 2 | 3 | 4
4 | 8 | 16 | 22 | 34

start

key = 8

end

$$\text{mid} = \frac{\text{start} + \text{end}}{2}$$

$$\text{mid} = \frac{\text{start} + \text{end}}{2}$$

$\rightarrow \text{if} (\text{arr}[\text{mid}] == \text{key})$
 return mid;

16 > 8

$\rightarrow \text{if} (\text{arr}[\text{mid}] > \text{key})$
 $\text{end} = \text{mid} - 1$

$$\frac{0+4}{2} = 2$$

→ if (arr[mid] > key) {
end = mid - 1;

Questions

23 May 2026

12:06

Q.1

main.cpp



Share

```
1 // Online C++ compiler to run C++ program online
2 #include <iostream>
3 using namespace std;
4 int main() {
5     int x = 5;
6     if(x == 5)
7         cout<<"X is equal to 5"<<endl;
8
9     return 0;
10 }
```

Q.2

main.cpp



Share

Run

```
1 // Online C++ compiler to run C++ program online
2 #include <iostream>
3 using namespace std;
4 int main() {
5     int x = 5;
6     if(x == 4);
7         cout<<"X is equal to 5"<<endl;
8
9     return 0;
10 }
```

Q.3.

main.cpp



Share

Run

```
1 // Online C++ compiler to run C++ program online
2 #include <iostream>
3 using namespace std;
4 int main() {
5     int x = 3;
6     if(x == 5)
7         cout<<"X is equal to 5"<<endl;
8     else
9         cout<<"X is not equal to 5"<<endl;
10    return 0;
11 }
```


0.4.

main.cpp



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Run

```
1 // Online C++ compiler to run C++ program online
2 #include <iostream>
3 using namespace std;
4 int main() {
5     float x = 0.5;
6     if(x == 0.5)
7         cout<<"abc"<<endl;
8     else
9         cout<<"def"<<endl;
10    return 0;
11 }
```

**What is array,
initialization, printing
array and so on.,**

Loops

23 May 2026

12:33

For loops,
While loops,
Do-while loops.

Questions 2

23 May 2026 12:19

Q.1

```
1 // Online C++ compiler to run C++ program online
2 #include <iostream>
3 using namespace std;
4 int main() {
5     int arr[5] = {1,2,3,4,5};
6     cout<<arr[0];
7
8     return 0;
9 }
```

Q.2

```
cpp
#include <iostream>
using namespace std;
int main() {
    int arr[5] = {1, 2, 3, 4, 5};
    for (int i = 0; i < 5; i++) {
        cout << arr[i] << " ";
    }
    return 0;
}
```

Q.3

```
cpp
#include <iostream>
using namespace std;
int main() {
    int arr[5] = {1, 2, 3, 4, 5};
    for (int i = 4; i >= 0; i--) {
        cout << arr[i] << " ";
    }
    return 0;
}
```

Q.5

```
cpp
#include <iostream>
using namespace std;
int main() {
    int arr[5] = {10, 20};
    for (int i = 0; i < 5; i++) {
        cout << arr[i] << " ";
    }
    return 0;
}
```

Q.6.

```
cpp
#include <iostream>
using namespace std;
int main() {
    int arr[5] = {1, 2, 3, 4, 5};
    int sum = 0;
    for (int i = 0; i < 5; i++) {
        sum += arr[i];
    }
    cout << sum;
    return 0;
}
```

Q.7.

```
cpp
#include <iostream>
using namespace std;
int main() {
    int arr[5] = {1, 2, 3, 4, 5};
    for (int i = 0; i < 5; i++) {
        arr[i] = arr[i] * 2;
    }
    for (int i = 0; i < 5; i++) {
        cout << arr[i] << " ";
    }
    return 0;
}
```